

(a) Answer quality

(b) Abstention behavior

	BERTScore F1	Correctness (0–2)	Faithfulness	Answer relevance	Refusal rate	Over-refusal rate
Vanilla base	0.728	0.740	N/A	0.762	1.6%	0.0%
SOTA base	0.724	1.199	N/A	0.839	0.0%	0.0%
BM25	0.732	0.122	0.651	0.240	86.2%	0.4%
ASR-only RAG	0.734	0.142	0.615	0.255	83.2%	0.4%
Vision-only RAG	0.734	0.126	0.627	0.250	84.4%	1.2%
Vector-only RAG	0.736	0.134	0.626	0.266	83.0%	0.4%
Graph-RAG	0.734	0.144	0.632	0.250	82.4%	0.4%